

SYSTEM CAN PROGRAMMING

To experiment system calls using C

```
santhiyashree@localhost:~$ touch fork_program.c
santhiyashree@localhost:~$ ls
emp.dat      fork_program  marks.awk    marks.dat
f1.txt       fork_program.c  santhiyashree241801249
emp.awk      fork_program.c
santhiyashree@localhost:~$ vi fork_program.c
santhiyashree@localhost:~$ gcc fork_program.c -o fork_program
santhiyashree@localhost:~$ ./fork_program
THIS LINE IS EXECUTED TWICE

PARENT PROCESS
Parent process ID:3851
Parent's process ID:3063
IT CAN BE EXECUTED TWICE
THIS LINE IS EXECUTED TWICE

CHILD PROCESS
Child process ID:3852
Parent process ID:2356
IT CAN BE EXECUTED TWICE
santhiyashree@localhost:~$
```

1. Create a process:

• Call the fork() function to create new process. Store the return value in a variable.

• If fork() returns:

- Forking failed
- 0: process is the child
- Positive integer: process is parent

```
rashiga241801225@localhost:~ — vi fork_program.c
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
int main(){
    int pid;
    pid = fork()
    printf("THIS LINE EXECUTED TWICE\n");
    if(pid == -1){
        printf("CHILD PROCESS NOT CREATED\n");
    }
    if(pid == 0){
        printf("CHILD PROCESS NOT CREATED\n");
        exit(0);
    }
    if(pid == 0){
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child process ID: %d\n",getpid());
        printf("Parent process ID: %d\n",getpid());
    }
}

"fork_program.c" 20L, 531B
```



```
rashiga241801225@localhost:~  
[rashiga241801225@localhost ~]$ touch fork_program.c  
[rashiga241801225@localhost ~]$ ls  
asd Desktop goa Music scripts third.sh  
back.sh Documents leapyear.sh Pictures second.sh Videos  
backup.c Downloads main.sh psg shell  
backup.sh fivith.sh marks.awk Public shell_lab  
backuup.sh fork_program.c marks.dat reverse.sh Templates  
delhi four.sh mumbai sample test  
[rashiga241801225@localhost ~]$ vi fork_program.c  
[rashiga241801225@localhost ~]$
```